

Heng Ge

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EDUCATION

UNIVERSITY OF FLORIDA

Aug 2023 - Now

Doctor of Philosophy in Biostatistics, GPA: 3.72/4.0

Gainesville, FL

- **Advisor:** Dr. Qing Lu
- **Dissertation:** Kernel Neural Network for Complex Genetic Risk Analysis
- **Related Courses:** Biostatistical Advanced Methods I & Biostatistical Advanced Methods II & Large Sample Theory

UNIVERSITY OF FLORIDA

Aug 2021 - May 2023

Master of Science in Biostatistics, GPA: 3.94/4.0

Gainesville, FL

- **Related Courses:** Biostatistical Methods I & Biostatistical Methods II & Introduction to Biostatistical Theory & Bayesian Biostats Methods & Biostatistical Computing & Applied Survival Analysis

SHANDONG UNIVERSITY

Sep 2016 - Jun 2020

Bachelor of Science in Statistics, GPA: 87/100

Shandong, China

- **Related Courses:** Mathematical Analysis & Advanced Algebra & Probability Theory & Mathematical Statistics & Time Series Analysis & Application of Regression Analysis & Multivariate Statistical Analysis & Basic Stochastic Processes & Real Function Theory & Functional Analysis & Partial Differential Equation

SHANDONG UNIVERSITY

Sep 2016 - Jun 2020

Bachelor of Economics in Finance (Double Major), GPA: 86/100

Shandong, China

- **Related Courses:** Microeconomics & Macroeconomics & Monetary Economics & Central Banking

RESEARCH INTEREST

My research focuses on developing statistical and computational methods for genetic and genomic studies, particularly in the context of high-dimensional and non-linear data. I am especially interested in improving association testing methods by integrating machine learning frameworks, such as deep learning and kernel-based approaches, with classical statistical models. My work addresses challenges arising in genome-wide association studies (GWAS), whole genome sequencing, and functional genomic assays such as bulk and single-cell RNA sequencing. Through novel method development and application to large-scale datasets like Alzheimer's Disease Neuroimaging Initiative (ADNI) and Religious Orders Study and Memory and Aging Project (ROSMAP), I aim to uncover complex genotype-phenotype relationships and advance our understanding of the genetic architecture underlying human diseases and related quantitative traits.

PUBLICATIONS & UNDER REVISION PAPERS

- [1]Chen, X., Wang, L., Hang, Y., **Ge, H.**, & Zha, H. (2019). Infinite-horizon Off-Policy Policy Evaluation with Multiple Behavior Policies. *International Conference on Learning Representations (ICLR 2020)*.
- [2]**Ge, H.** & Lu, Q.. (2026). Neural Tangent Kernels for Complex Genetic Risk Prediction: Bridging Deep Learning and Kernel Methods in Genomics. *Under review at Briefings in Bioinformatics*.
- [3]**Ge, H.**, Xue, W., Wu, M. & Lu, Q.. (2026). DiffKnock: Diffusion-based Knockoff Statistics for Neural Networks Inference. *Under review at Briefings in Bioinformatics*.
- [4]**Ge, H.**, Shen, X., Hou, T., Xue, W., Chen, L. & Lu, Q.. (2025). A Penalized Kernel Neural Network For Complex Phenotype Prediction. *Under review at the Statistics in Medicine*.
- [5]Shen, X., Tong, X., **Ge, H.**, Xue, W., Liang, M. & Lu, Q.. (2026). An Efficient Kernel-Based Neural Network for High-dimensional Genomic Data Analysis. *Under review at Biometrics*.
- [6]Hou, T., Jiang, C., **Ge, H.**, Xue, W., Chen, L. & Lu, Q.. (2026). Kernel-Based Neural Network Tests for High-dimensional Genetic Association Analysis *Under review at the Annals of Applied Statistics*.
- [7]**Ge, H.**, Shen, X. & Lu, Q.. (2026). A Generalized Penalized Kernel Neural Network For Complex Phenotype Prediction. *Ready to submit*.
- [8]**Ge, H.**, Hou, T. & Lu, Q.. (2026). An Association Test for Sequencing Data Based on Penalized Kernel Neural Network. *Ready to submit*.
- [9]**Ge, H.**, Hou, T. & Lu, Q.. (2026). An Association Test for Sequencing Data Based on Penalized Kernel Neural Network for Multivariate Phenotype. *Ready to submit*.

CONSULTING EXPERIENCE

Breathing Research and Therapeutics Center

Nov 2025 - Now

Graduate Research Assistant

Gainesville, FL

- Led end-to-end statistical analysis for a double-blind, placebo-controlled, randomized-order, repeated-measures cross-over clinical trial evaluating dAIH, respiratory strength training (RST), and their combination.
- Designed and implemented a linear mixed-effects modeling framework (random intercept for participant) to evaluate treatment effects on respiratory endpoints (MIP, MEP, FVC, P0.1), including omnibus testing and multiplicity-adjusted planned contrasts (Holm correction).
- Conducted sensitivity analyses comparing full, reduced, and first-block models; formally assessed period effects (Block 1 vs Blocks 2–4) and performed backward selection to evaluate covariate stability and model robustness.
- Generated reproducible analysis pipelines and clinician-facing reports with trajectory visualization, change-from-baseline metrics, and structured hypothesis testing outputs.

TEACHING EXPERIENCE

PHC6099: Programming Basics for Biostatistics

Fall 2026

Instructor

Gainesville, FL

PHC6937: Special Topics in Public Health

Fall 2026

Teaching Assistant

Gainesville, FL

INDUSTRY EXPERIENCE

Vipshop Holdings Ltd

Jul 2020 - May 2021

Data Scientist

Guangzhou, China

- Developed and implemented machine learning models to predict customer lifetime value, improving targeting accuracy and reducing customer acquisition costs.
- Built automated fraud detection framework using detection algorithms and rule-based systems to identify fraudulent clicks and ineffective ad placements.

HONORS & AWARDS

PhD Student Travel Award

Mar 2026

Biostatistics departmental award supporting travel expenses for PhD student presenters at meetings.

AI Fellow Honorable Mention

Apr 2025

College award designed to build national recognition of AI innovation within the college's doctoral programs.

Outstanding Master Student Award

May 2023

Highest cumulative GPA through the semesters prior to graduation among all Biostatistics Master students.

Honorable Mention of MCM/ICM

Jan 2019

Top 15% in 25370 mathematical modelling teams from all over the world.

First Class Award of the College Students Mathematical Modeling Competition

Oct 2018

Top 5% in 3549 mathematical modelling teams in Shandong province.

PRESENTATIONS

2023 ASA Annual Florida Chapter Meetings

Mar 2023

Talk: Selection of Variance Component for Kernel Neural Network

Service: Organizer and Session Chair of Student Contributed Session A

2023 ENAR Spring Meetings

March 2023

Poster: Selection of Variance Component for Kernel Neural Network

2024 ASA Joint Statistical Meetings

Aug 2024

Talk: A Penalized Kernel Neural Network for Complex Genetic Risk Prediction Analysis

Center for Genetic Epidemiology and Bioinformatics: Ongoing Research Seminar

Jan 2025

Talk: A Penalized Kernel Neural Network for Complex Genetic Risk Analysis

2025 MCBIOS Conference

Mar 2025

Poster: An Association Test for Sequencing Data Based on Penalized Kernel Neural Network

2025 PHHP Research Day

Apr 2025

Talk: An Association Test for Sequencing Data Based on Penalized Kernel Neural Network

2026 MCBIOS Conference

Mar 2026

Invited Talk: Neural Tangent Kernels for Nonlinear Genetic Risk Prediction: Bridging Deep Learning and Statistical Genetics
2026 ASA Joint Statistical Meetings Aug 2026
Talk: Neural Tangent Kernels for Nonlinear Genetic Risk Prediction: Bridging Deep Learning and Statistical Genetics
Service: Session Chair of Hypothesis Testing and AI-Assisted Inference

REVIEW SERVICE

Society for Artificial Intelligence and Statistics	Nov 2025
International Conference on Machine Learning	Feb 2026
Statistics Innovation	June 2026

MEMBERSHIP

American Statistical Association (ASA)	2023 - Now
Eastern North American Region International Biometric Society (ENAR)	2023 - Now
Midsouth Computational Biology and Bioinformatics Society (MCBIOS)	2025 - Now

SKILLS

Programming Languages: Python, R, MATLAB, SAS, C/C++
Bioinformatics & Data Science: Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib, Seurat, ggplot2, dplyr
Machine Learning & AI: Kernel Methods, Transformer, Diffusion, Transfer Learning, Predictive Modeling, Feature Engineering
Single-cell Bioinformatics: scRNA-seq (Seurat), Transcriptomic Data Processing, Dimensionality Reduction, Clustering
Operating Systems: Windows, Linux
Languages: English, Chinese